Mechanics 1

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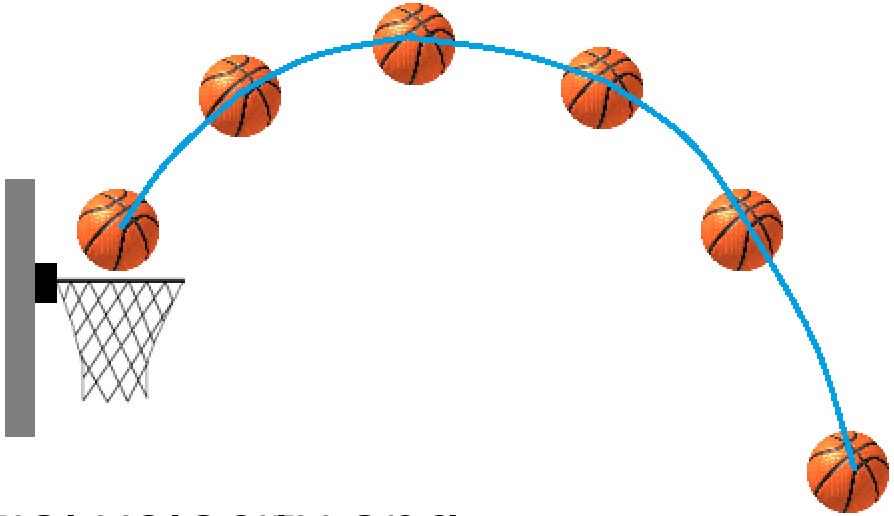
1.4 Projectile motion



- A projectile is any body that is given an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration and air resistance.
- A batted baseball, a thrown football, a package dropped from an airplane, and a bullet shot from a rifle are all projectiles.
- The path followed by a projectile is called its trajectory

Assumptions for projectile motion:

1. **Free-fall acceleration**

   On Earth, every object experience an acceleration that has magnitude $g = 9.80\,\mathrm{m\,s^{-2}}$

   and is a constant. The direction is vertically towards the ground the negative-y direction

2. **Neglect air resistance**

   Air resistance will cause additional acceleration in other directions. For now, we shall neglect the effects of air resistance.

Taking both assumptions into account, the acceleration components of a projectile is $a_x = 0$, $a_y = -g$, and $a_z = 0$, and

$$\begin{aligned}\vec{a} &= a_x\,\hat{\imath} + a_y\,\hat{\jmath} + a_z\,\hat{k} \\ &= -g\,\hat{\jmath}.\end{aligned}$$

the initial velocity at $t = 0$ $\vec{v} = v_0 \cos\theta\,\hat{\imath} + v_0 \sin\theta\,\hat{\jmath},$

and the initial position of the projectile to be at the origin. In other words, at $t = 0$, we have $x = 0$ and $y = 0$ (or $\vec{r}_0 = \vec{0}$).

By the definition of acceleration, $\frac{d\vec{v}}{dt}$, and since g is a constant,

$$\vec{v} - \vec{v}_0 = \int_0^t \vec{a}\, \mathrm{d}t = -\int_0^t g\,\hat{\jmath}\, \mathrm{d}t = -gt\,\hat{\jmath}$$

$$\vec{v} = \vec{v}_0 - gt\ \hat{\jmath} = (v_0 \cos\theta)\ \hat{\imath} + (v_0 \sin\theta)\ \hat{\jmath} - gt\ \hat{\jmath}$$

$$\vec{v} = (v_0 \cos\theta)\ \hat{\imath} + (v_0 \sin\theta - gt)\ \hat{\jmath}$$

By the definition of velocity $\frac{d\vec{r}}{dt} = \vec{v}$, we have

$$\vec{r} - \vec{r}_0 = \int_0^t \vec{v}\, \mathrm{d}t = \int_0^t \left[(v_0 \cos\theta)\ \hat{\imath} + (v_0 \sin\theta - gt)\ \hat{\jmath}\right] \mathrm{d}t$$

$$\vec{r} = x\ \hat{\imath} + y\ \hat{\jmath} = (tv_0 \cos\theta)\ \hat{\imath} + \left(tv_0 \sin\theta - \frac{1}{2}gt^2\right)\ \hat{\jmath}.$$

Comparing the x- and y-components on both sides of the equation, we have

$$\boxed{x = tv_0 \cos\theta, \quad y = tv_0 \sin\theta - \frac{1}{2}gt^2.}$$

We can derive another useful equation by eliminating t.

$$t = \frac{x}{v_0 \cos\theta}$$

$$y = \left(\frac{x}{\cancel{v_0}\cos\theta}\right)\cancel{v_0}\sin\theta - \frac{1}{2}g\left(\frac{x}{v_0\cos\theta}\right)^2$$

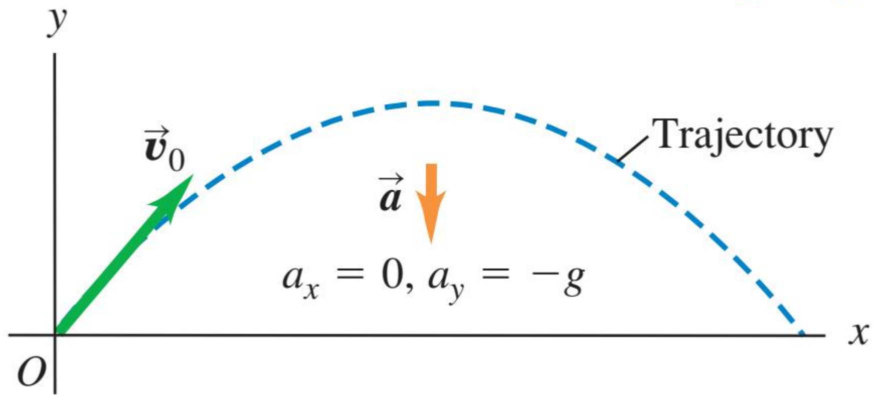
$$\boxed{y = (\tan\theta)\,x - \left(\frac{g}{2v_0^2\cos^2\theta}\right)x^2.}$$

projectile equation

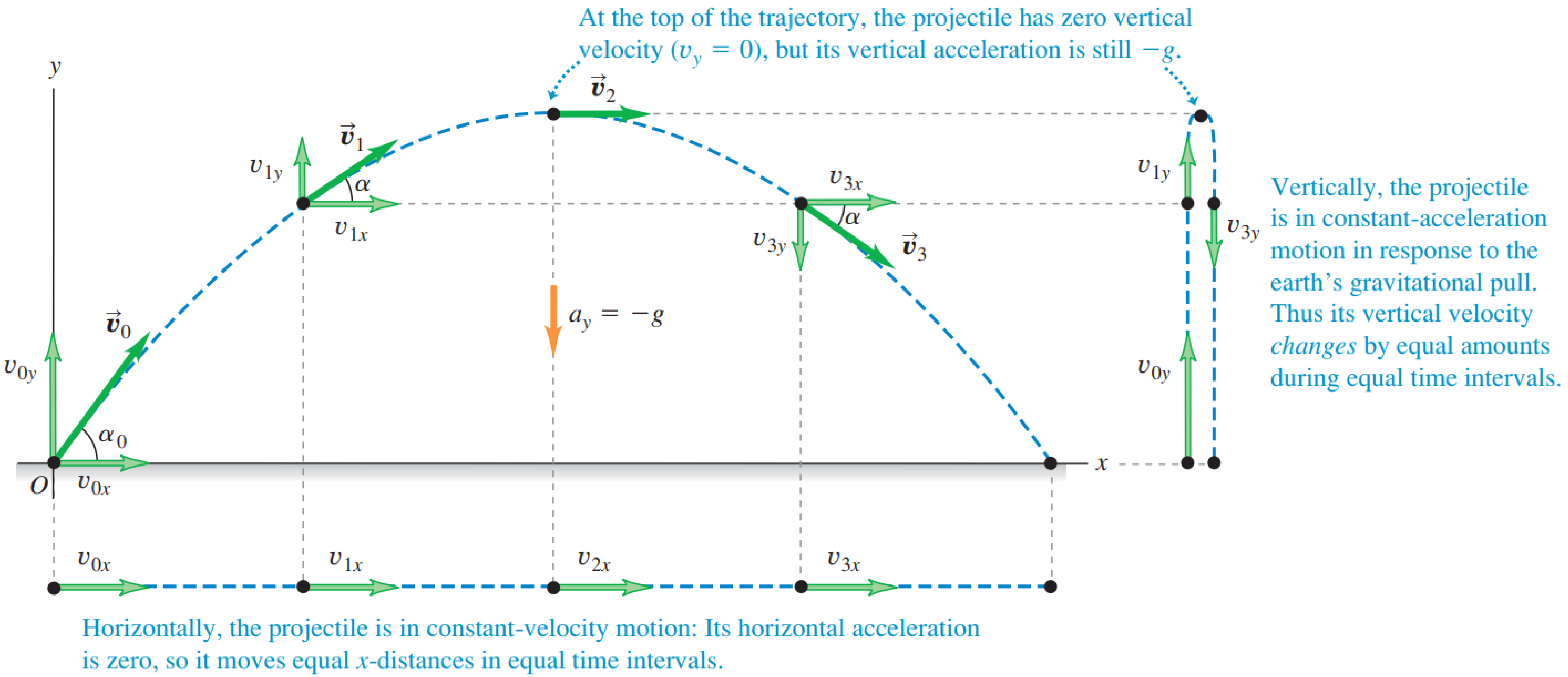
the trajectory describes a **parabola**

3.15 The trajectory of an idealized projectile.

- A projectile moves in a vertical plane that contains the initial velocity vector $\vec{v}_0$.
- Its trajectory depends only on $\vec{v}_0$ and on the downward acceleration due to gravity.



3.17 If air resistance is negligible, the trajectory of a projectile is a combination of horizontal motion with constant velocity and vertical motion with constant acceleration.



Height and range of a projectile I: A batted baseball

A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0\ \mathrm{m/s}$ at an angle $\alpha_0 = 53.1°$. (a) Find the position of the ball and its velocity (magnitude and direction) at $t = 2.00$ s. (b) Find the time when the ball reaches the highest point of its flight, and its height h at this time. (c) Find the *horizontal range R*—that is, the horizontal distance from the starting point to where the ball hits the ground.

Example 1.4.1:

A man stands on the roof of a 15-m tall building and throws a rock with a velocity of magnitude 30.0 $\mathrm{m\,s^{-1}}$ at an angle of 33° above the horizontal. Ignore air resistance, calculate

(a) The maximum height above the roof reached by the rock.

(b) Speed of the rock just before it hits the ground.

(c) The horizontal range from the base of the building to the point where the rock strikes the ground.

Example 1.4.2:

Firefighters are shooting a stream of water at a burning building using a high-pressure hose that shoots out water with a speed of 25 $\mathrm{m\,s^{-1}}$ as it leaves the end of the hose. Once it leaves the hose, the water moves in projectile motion. The firemen adjusts the angle of elevation α of the hose until the water takes 3.0 s to reach a building 45.0 m away. Ignore air resistance and assume that the end of the hose is at ground.

(a) Find α.

(b) Find the speed and acceleration of the water at the highest point.

(c) How high above the ground does the water strike the building?

Example 1.4.3:

A projectile is launched on flat ground with initial speed v_0 and angle θ above the horizontal.

(a) How far away from the starting point does it land on the ground?

(b) What is the initial angle θ such that the distance is furthest possible?

(a) *The projectile equation is given by Eq. (1.66). When it lands on the horizontal ground, $y = 0$. Therefore*

$$0 = x\tan\theta - \left(\frac{g}{2v_0^2\cos^2\theta}\right)x^2$$

$$0 = x\left[\frac{\sin\theta}{\cos\theta} - \frac{gx}{2v_0^2\cos^2\theta}\right]$$

$$x = 0 \quad or \quad \frac{gx}{2v_0^2\cos^{\cancel{2}}\theta} = \frac{\sin\theta}{\cancel{\cos\theta}}.$$

The first solution $x = 0$ is just the starting point. Therefore the relevant solution is the second one,

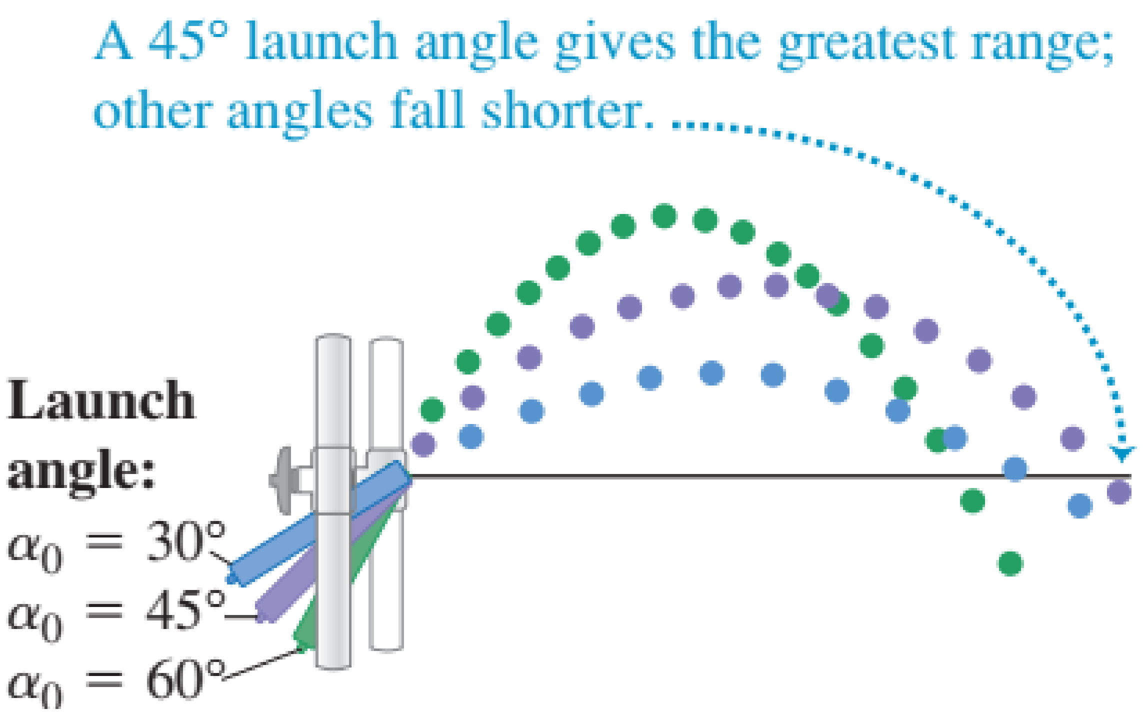
$$x = R = \frac{2v_0}{g}\sin\theta\cos\theta$$

$$R = \frac{v_0^2}{g}\sin 2\theta.$$

(b) *Finding the angle such that $R = \frac{v_0}{g}\sin 2\theta$ is the largest — this value is the largest when the sine function takes the highet possible number, which is 1. This occurs*

$$2\theta = \frac{\pi}{2} \quad \rightarrow \quad \theta = \frac{\pi}{4}.$$

3.24 A launch angle of 45° gives the maximum horizontal range. The range is shorter with launch angles of 30° and 60°.



1.5 Circular motion

The motion of objects moving on a circular path

1.5.1 Uniform circular motion

- An object is moving on a circular path also has constant speed at all times, it is called **uniform circular motion**
- However, since the path is circular, the direction is constantly changing. This means the velocity vector always has the same magnitude, but the direction is non-constant. A non-constant velocity vector implies that the object is experiencing acceleration.

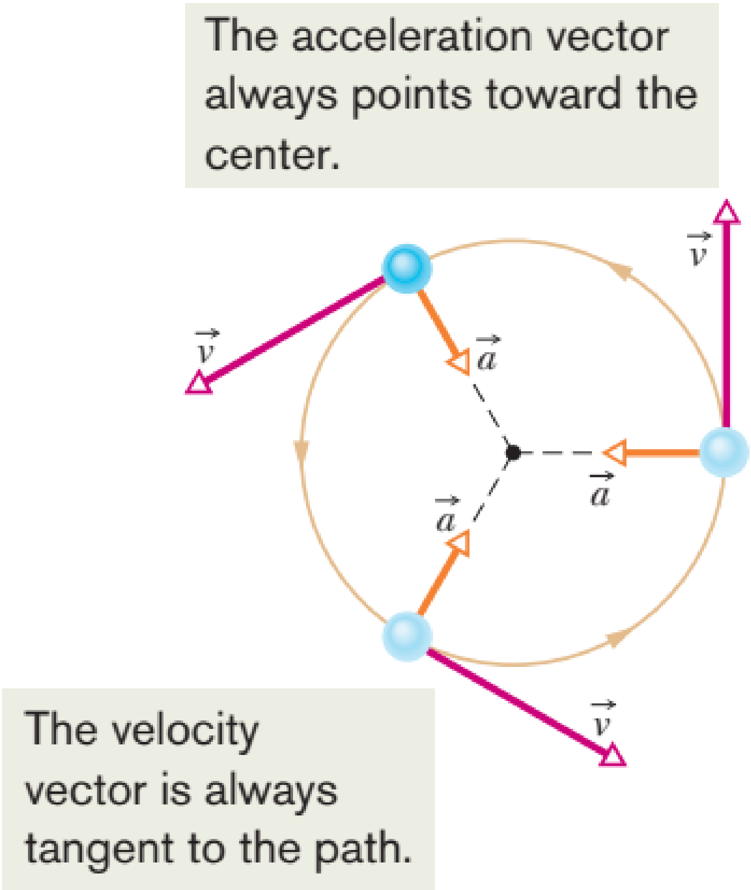
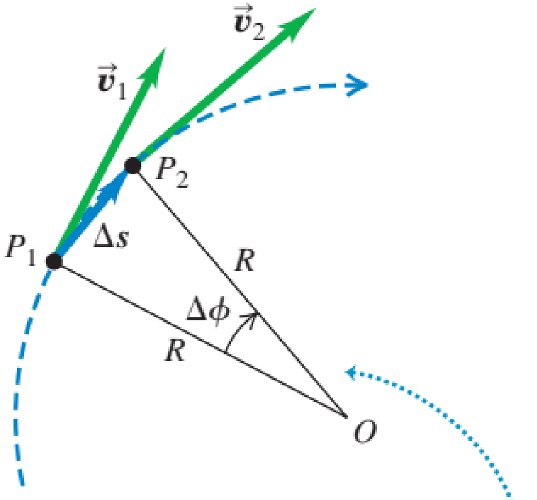


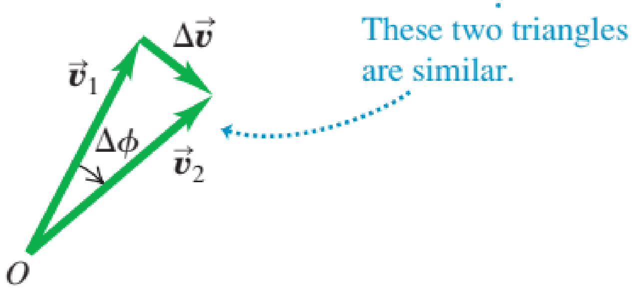
Figure 4-16 Velocity and acceleration vectors for uniform circular motion.

3.28 Finding the velocity change $\Delta\vec{v}$, average acceleration $\vec{a}_{av}$, and instantaneous acceleration $\vec{a}_{rad}$ for a particle moving in a circle with constant speed.

(a) A particle moves a distance Δs at constant speed along a circular path.



(b) The corresponding change in velocity and average acceleration



The angles labeled $\Delta\phi$ in Figs. 3.28a and 3.28b are the same because $\vec{v}_1$ is perpendicular to the line OP_1 and $\vec{v}_2$ is perpendicular to the line OP_2. Hence the triangles in Figs. 3.28a and 3.28b are *similar*. The ratios of corresponding sides of similar triangles are equal, so

$$\frac{|\Delta\vec{v}|}{v_1} = \frac{\Delta s}{R} \qquad \text{or} \qquad |\Delta\vec{v}| = \frac{v_1}{R}\Delta s$$

The magnitude a_{av} of the average acceleration during Δt is therefore

$$a_{av} = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{v_1}{R}\frac{\Delta s}{\Delta t}$$

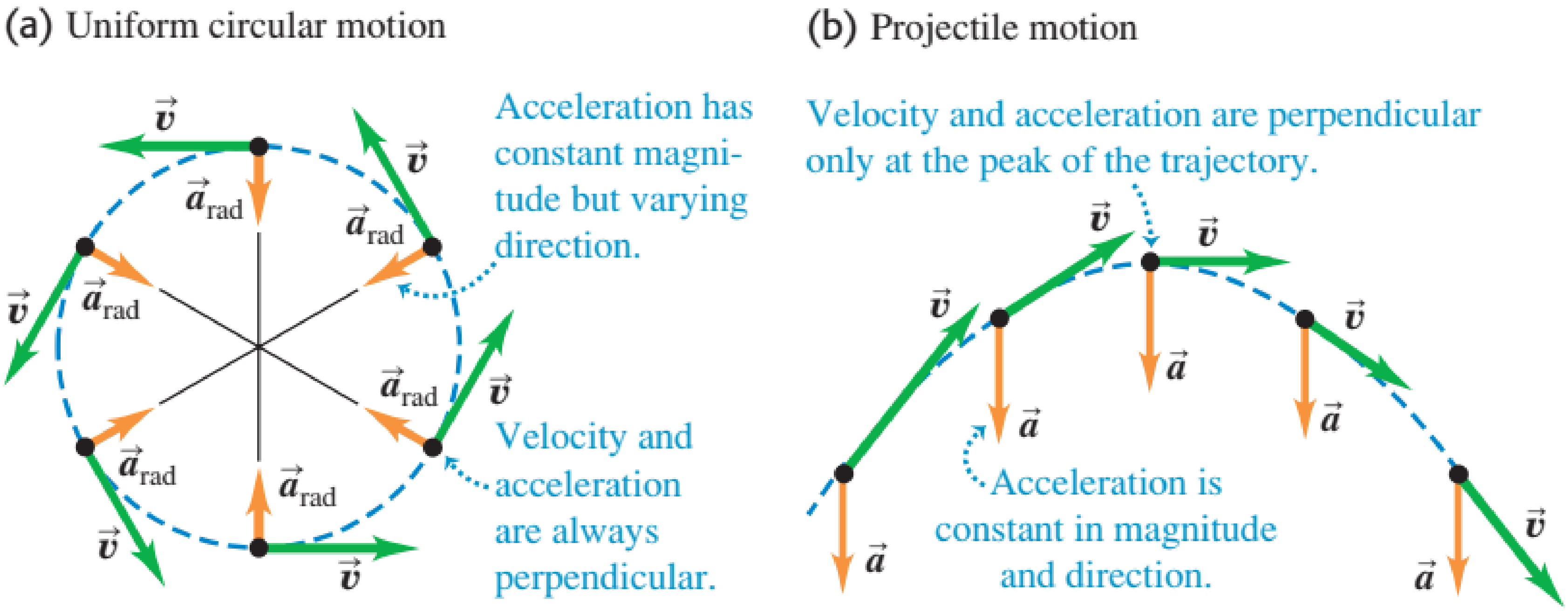
The magnitude a of the *instantaneous* acceleration $\vec{a}$ at point P_1 is the limit of this expression as we take point P_2 closer and closer to point P_1:

$$a = \lim_{\Delta t \to 0} \frac{v_1}{R}\frac{\Delta s}{\Delta t} = \frac{v_1}{R} \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t}$$

$$a_{rad} = \frac{v^2}{R} \qquad \text{(uniform circular motion)}$$

$$v = \frac{2\pi R}{T} \qquad a_{rad} = \frac{4\pi^2 R}{T^2} \qquad \text{(uniform circular motion)}$$

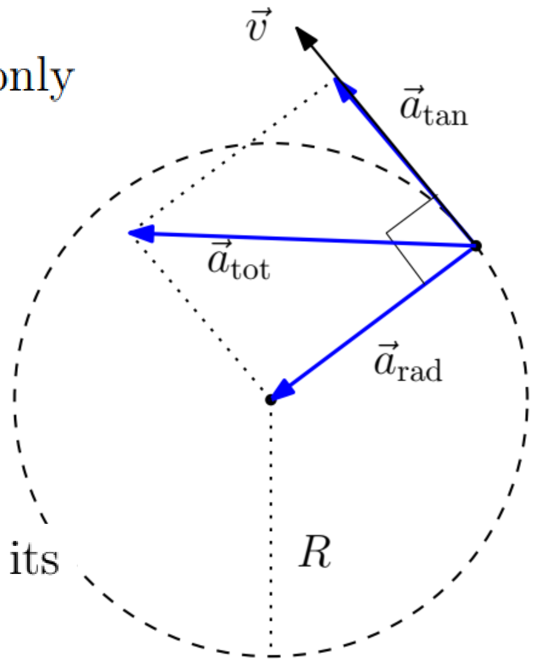
3.29 Acceleration and velocity (a) for a particle in uniform circular motion and (b) for a projectile with no air resistance.



1.5.2 Non-uniform circular motion

If the speed is not constant, there will also be a **tangential** acceleration

The vector $\vec{a}_{\text{tan}}$ is always parallel to the velocity vector $\vec{v}$ at any instant. Therefore it only changes its speed, not the direction.



The **acceleration** vector is given by

$$\vec{a}_{\text{tot}} = \vec{a}_{\text{rad}} + \vec{a}_{\text{tan}}$$

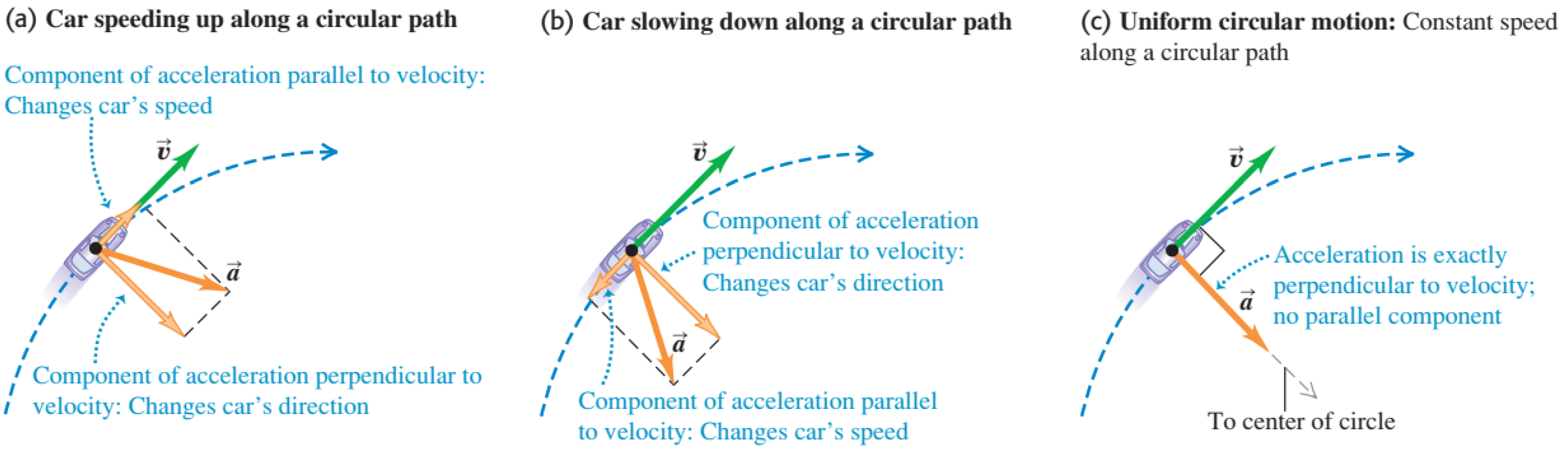
Tangential acceleration causes change in the particle's speed, but doesn't affect its direction. The magnitude is given by the particle's rate of change of speed:

$$|\vec{a}_{\text{tan}}| = a_{\text{tan}} = \frac{\mathrm{d}\,|\vec{v}|}{\mathrm{d}t}.$$

Radial acceleration causes change in the particle's direction only, and doesn't affect its speed. The magnitude is given by

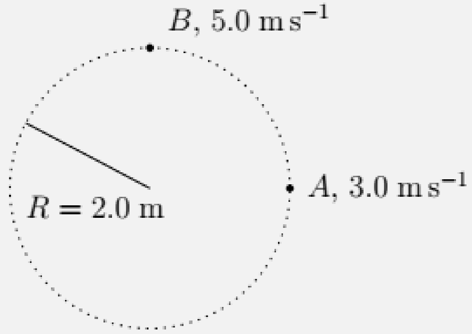
$$|\vec{a}_{\text{rad}}| = a_{\text{rad}} = \frac{v^2}{r}.$$

3.27 A car moving along a circular path. If the car is in uniform circular motion as in (c), the speed is constant and the acceleration is directed toward the center of the circular path (compare Fig. 3.12).



Example 1.5.1:

A particle is moving in circular motion of radius $R = 2.0$ m. Initially, the particle is at point A and has a speed of $3.0\ \mathrm{m\,s^{-1}}$. 0.1 seconds later, the particle reaches point B with speed $5.0\ \mathrm{m\,s^{-1}}$. The magnitude of tangential acceleration, a_{tan}, is constant.



(a) Calculate the value of a_{tan}.

(b) Calculate a_{rad} at point A and B, respectively.

(c) Sketch the vectors $\vec{a}_{\mathrm{rad}}$, $\vec{a}_{\mathrm{tan}}$ when the particle is at point B. Sketch also the total acceleration vector, $\vec{a}_{\mathrm{tot}}$.

(d) Find the angle between $\vec{a}_{\mathrm{tot}}$ and $\vec{a}_{\mathrm{tan}}$.

(a) Since $\vec{a}_{\tan}$ is constant, the magnitudes of tangential components obey the kinematic equations:

$$v = u + a_{\tan} t$$

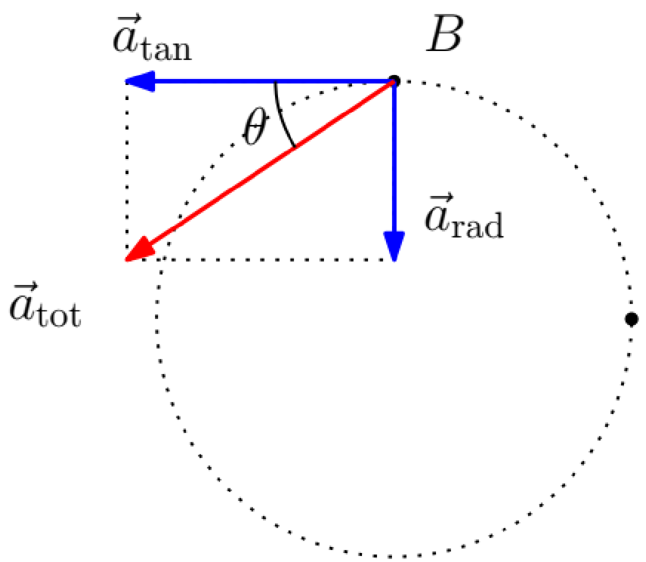
$$5.0 = 3.0 + a_{\tan}(4.0) \quad \rightarrow \quad a_{\tan} = \frac{5.0 - 3.0}{0.1} = 20 \text{ m s}^{-2}$$

(b) Using the formula $a_{\text{rad}} = \frac{v^2}{R}$

$$\textit{At point } A : a_{\text{rad}} = \frac{v_A^2}{R} = \frac{3.0^2}{2.0} = 4.5 \text{ m s}^{-2},$$

$$\textit{At point } B : a_{\text{rad}} = \frac{v_B^2}{R} = \frac{5.0^2}{2.0} = 12.5 \text{ m s}^{-2}$$

(c) At point B,



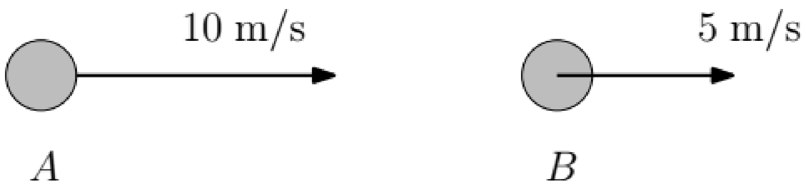
(d) The angle between $\vec{a}_{\text{tot}}$ and $\vec{a}_{\tan}$ can be calculated using

$$\tan\theta = \frac{a_{\text{rad}}}{a_{\tan}} = \frac{12.5}{20} \quad \rightarrow \quad \theta = \tan^{-1}(0.65) = 32°.$$

1.6 Relative motion

Consider two cars, A and B. Car A moves at $v_A = 10.0\ \mathrm{ms^{-1}}$, and car B moves at $v_B = 5.0\ \mathrm{ms^{-1}}$.

- From Car A's perspective, what is the velocity of car B?
- From Car B's perspective, what is the velocity of car A?

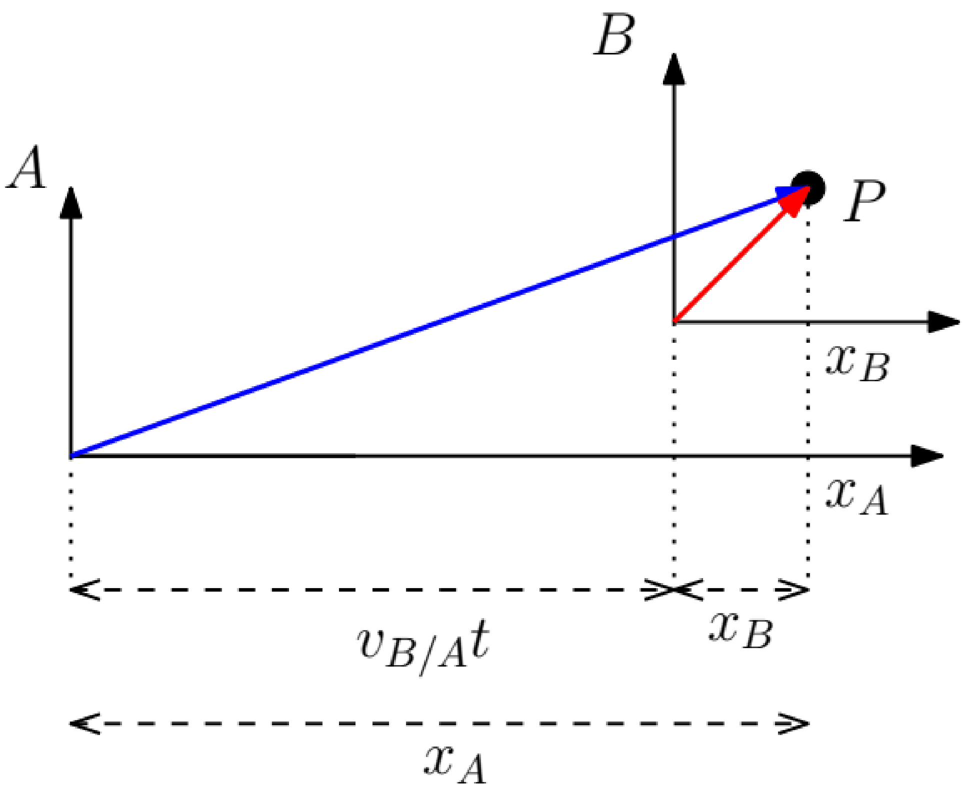


$v_{A/B}$ to mean 'velocity of A as observed by B'.

- Car A observes B approaching it at $-5.0\ \mathrm{ms^{-1}}$. ($v_{B/A} = v_B - v_A$.)
- Car B observes A approaching it at $+5.0\ \mathrm{ms^{-1}}$. ($v_{A/B} = v_A - v_B$.)

Suppose Cars *A* and *B* use their own separate coordinate system. Each of them taking themselves as the origin.

Inertial frames: Coordinate systems which are moving at constant velocity relative to each other.



Car A sees Car B moving away from her at $v_{B/A}$.

Initially, at $t = 0$, both cars start at the same position.

After time t, the distance between A and B becomes

$$v_{B/A}t$$

They both observe the same point P

In A's coordinate system, the x-position of P is x_A. In B's coordinate system, the position of P is at x_B.

$$x_{P/A} = x_{P/B} + v_{B/A}t.$$

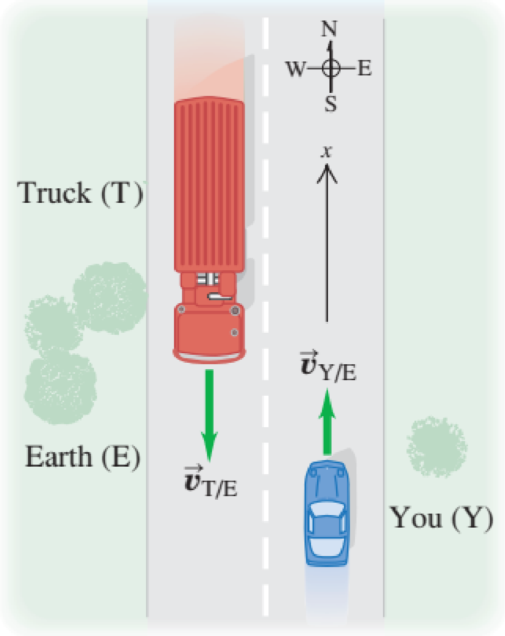
Differentiating both sides,

$$v_{P/A} = v_{P/B} + v_{B/A}.$$

Example 3.13 Relative velocity on a straight road

You drive north on a straight two-lane road at a constant 88 km/h. A truck in the other lane approaches you at a constant 104 km/h (Fig. 3.33). Find (a) the truck's velocity relative to you and (b) your velocity relative to the truck. (c) How do the relative velocities change after you and the truck pass each other? Treat this as a one-dimensional problem.

3.33 Reference frames for you and the truck.



SOLUTION

IDENTIFY and SET UP: In this problem about relative velocities along a line, there are three reference frames: you (Y), the truck (T), and the earth's surface (E). Let the positive x-direction be north (Fig. 3.33). Then your x-velocity relative to the earth is $v_{Y/E\text{-}x} = +88$ km/h. The truck is initially approaching you, so it is moving south and its x-velocity with respect to the earth is $v_{T/E\text{-}x} = -104$ km/h. The target variables in parts (a) and (b) are $v_{T/Y\text{-}x}$ and $v_{Y/T\text{-}x}$, respectively. We'll use Eq. (3.33) to find the first target variable and Eq. (3.34) to find the second.

EXECUTE: (a) To find $v_{T/Y\text{-}x}$, we write Eq. (3.33) for the known $v_{T/E\text{-}x}$ and rearrange:

$$v_{T/E\text{-}x} = v_{T/Y\text{-}x} + v_{Y/E\text{-}x}$$
$$v_{T/Y\text{-}x} = v_{T/E\text{-}x} - v_{Y/E\text{-}x}$$
$$= -104 \text{ km/h} - 88 \text{ km/h} = -192 \text{ km/h}$$

The truck is moving at 192 km/h in the negative x-direction (south) relative to you.

(b) From Eq. (3.34),

$$v_{Y/T\text{-}x} = -v_{T/Y\text{-}x} = -(-192 \text{ km/h}) = +192 \text{ km/h}$$

You are moving at 192 km/h in the positive x-direction (north) relative to the truck.

(c) The relative velocities do *not* change after you and the truck pass each other. The relative *positions* of the bodies don't matter. After it passes you the truck is still moving at 192 km/h toward the south relative to you, even though it is now moving away from you instead of toward you.

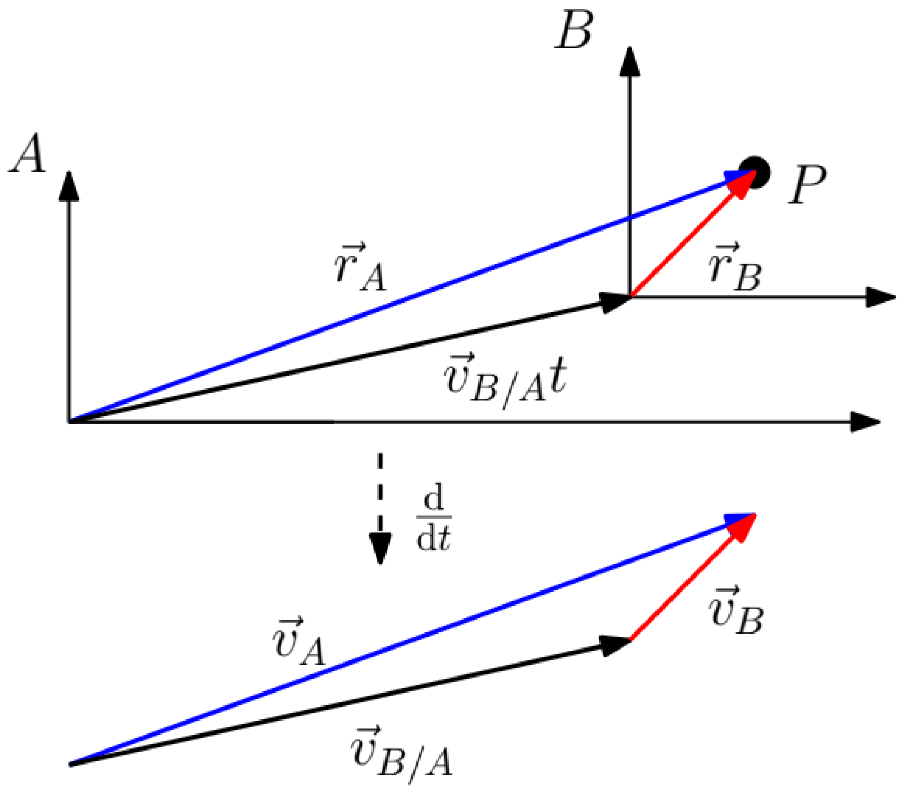
EVALUATE: To check your answer in part (b), use Eq. (3.33) directly in the form $v_{Y/T\text{-}x} = v_{Y/E\text{-}x} + v_{E/T\text{-}x}$. (The x-velocity of the earth with respect to the truck is the opposite of the x-velocity of the truck with respect to the earth: $v_{E/T\text{-}x} = -v_{T/E\text{-}x}$.) Do you get the same result?

Generalization to 2D and 3D

$$\vec{r}_{P/A} = \vec{r}_{P/B} + \vec{v}_{B/A}t.$$

Differentiating both sides,

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}.$$



Relative motion in 2D.

Example 3.14 Flying in a crosswind

An airplane's compass indicates that it is headed due north, and its airspeed indicator shows that it is moving through the air at 240 km/h. If there is a 100-km/h wind from west to east, what is the velocity of the airplane relative to the earth?

SOLUTION

IDENTIFY and SET UP: This problem involves velocities in two dimensions (northward and eastward), so it is a relative velocity problem using vectors. We are given the magnitude and direction of the velocity of the plane (P) relative to the air (A). We are also given the magnitude and direction of the wind velocity, which is the velocity of the air A with respect to the earth (E):

$$\vec{v}_{P/A} = 240 \text{ km/h} \qquad \text{due north}$$
$$\vec{v}_{A/E} = 100 \text{ km/h} \qquad \text{due east}$$

We'll use Eq. (3.36) to find our target variables: the magnitude and direction of the velocity $\vec{v}_{P/E}$ of the plane relative to the earth.

EXECUTE: From Eq. (3.36) we have

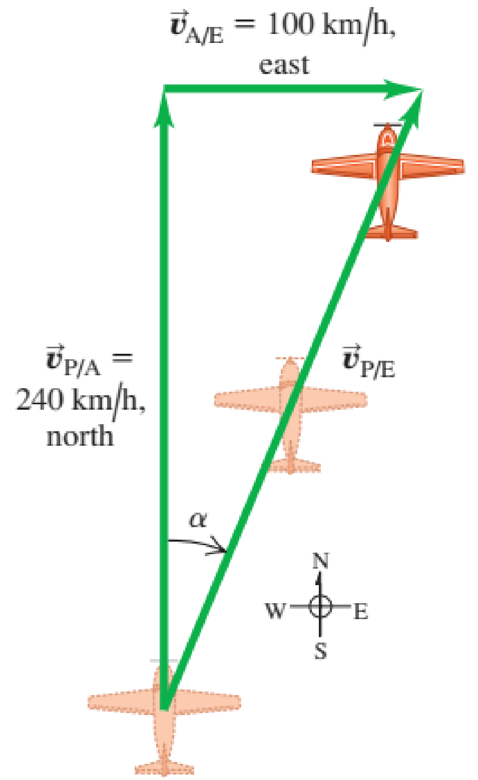
$$\vec{v}_{P/E} = \vec{v}_{P/A} + \vec{v}_{A/E}$$

Figure 3.35 shows that the three relative velocities constitute a right-triangle vector addition; the unknowns are the speed $v_{P/E}$ and the angle α. We find

$$v_{P/E} = \sqrt{(240 \text{ km/h})^2 + (100 \text{ km/h})^2} = 260 \text{ km/h}$$
$$\alpha = \arctan\left(\frac{100 \text{ km/h}}{240 \text{ km/h}}\right) = 23^\circ \text{ E of N}$$

EVALUATE: You can check the results by taking measurements on the scale drawing in Fig. 3.35. The crosswind increases the speed of the airplane relative to the earth, but pushes the airplane off course.

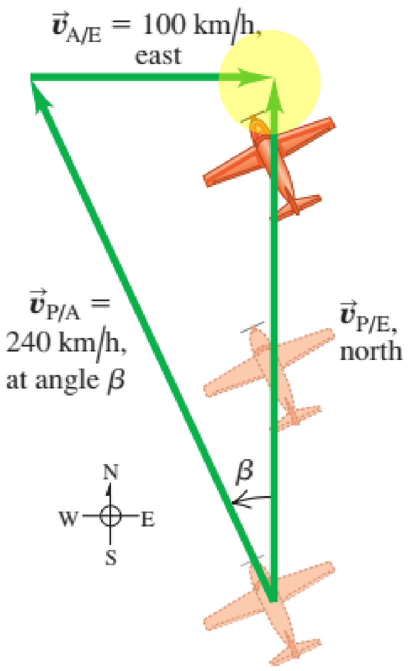
3.35 The plane is pointed north, but the wind blows east, giving the resultant velocity $\vec{v}_{P/E}$ relative to the earth.



Example 3.15 Correcting for a crosswind

With wind and airspeed as in Example 3.14, in what direction should the pilot head to travel due north? What will be her velocity relative to the earth?

3.36 The pilot must point the plane in the direction of the vector $\vec{v}_{P/A}$ to travel due north relative to the earth.



SOLUTION

IDENTIFY and SET UP: Like Example 3.14, this is a relative velocity problem with vectors. Figure 3.36 is a scale drawing of the situation. Again the vectors add in accordance with Eq. (3.36) and form a right triangle:

$$\vec{v}_{P/E} = \vec{v}_{P/A} + \vec{v}_{A/E}$$

As Fig. 3.36 shows, the pilot points the nose of the airplane at an angle β into the wind to compensate for the crosswind. This angle, which tells us the direction of the vector $\vec{v}_{P/A}$ (the velocity of the airplane relative to the air), is one of our target variables. The other target variable is the speed of the airplane over the ground, which is the magnitude of the vector $\vec{v}_{P/E}$ (the velocity of the airplane relative to the earth). The known and unknown quantities are

$$\begin{aligned} \vec{v}_{P/E} &= \text{magnitude unknown} && \text{due north} \\ \vec{v}_{P/A} &= 240\ \text{km/h} && \text{direction unknown} \\ \vec{v}_{A/E} &= 100\ \text{km/h} && \text{due east} \end{aligned}$$

EXECUTE: From Fig. 3.36 the speed $v_{P/E}$ and the angle β are

$$v_{P/E} = \sqrt{(240\ \text{km/h})^2 - (100\ \text{km/h})^2} = 218\ \text{km/h}$$

$$\beta = \arcsin\left(\frac{100\ \text{km/h}}{240\ \text{km/h}}\right) = 25°$$

The pilot should point the airplane 25° west of north, and her ground speed is then 218 km/h.

EVALUATE: There were two target variables—the magnitude of a vector and the direction of a vector—in both this example and Example 3.14. In Example 3.14 the magnitude and direction referred to the *same* vector ($\vec{v}_{P/E}$); here they refer to *different* vectors ($\vec{v}_{P/E}$ and $\vec{v}_{P/A}$).

While we expect a *headwind* to reduce an airplane's speed relative to the ground, this example shows that a *crosswind* does, too. That's an unfortunate fact of aeronautical life.